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LATE PRECAMBRIAN AND LOWER PALAEOZOIC PALAEOMAGNETIC RESULTS FROM SOUTH AUSTRALIA AND WESTERN AUSTRALIA

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The late Precambrian to Lower Palaeozoic apparent polar wander curve previously defined from palaeomagnetic studies in central and northern Australia has now been shown to be valid for southern and western Australia. During the latest Precambrian and Lower Palaeozoic a pole path of length 180° is now identified. In the northern Flinders Ranges, South Australia, palaeomagnetic poles have been obtained from the late Precambrian Pound Quartzite (60°S, 6°E, $N = 10$, $A_{95} = 23.5°$), Lower Cambrian sediments (36°S, 33°E; $N = 11$, $A_{95} = 16.5°$) and the Middle Cambrian–?Lower Ordovician Lake Frome Group (16°S, 25°E; $N = 20$, $A_{95} = 12.5°$). From Western Australia the Tumblagooda Sandstone of probable Ordovician age yields a palaeomagnetic pole at 30°S, 31°E ($N = 17$, $A_{95} = 9°$). Reappraisal of some previous studies on the Dundas Group of Tasmania and on bore cores from the Georgina Basin (northern Australia) and Yorke Peninsula (South Australia) indicates those results are compatible with the data presented.

1. Introduction

Since the early work of Irving and Green [1] on the Palaeozoic of northern Australia and southeastern Australia subsequent investigations in those regions [2–7] and in central Australia [8,9] have led to a fuller appreciation of their spatial relationships to one another during late Precambrian and Palaeozoic time [10]. The purpose of the present study is to assess whether palaeomagnetic results from South Australia and Western Australia also match the common polar wander curve previously identified from central and northern Australia [11].

Although we report the first palaeomagnetic results from Western Australia, Briden has previously collected from late Precambrian and Lower Palaeozoic formations in South Australia [12–14]. Unfortunately, he was unable positively to identify a primary magnetization and concluded that during the Early Tertiary, remagnetization had occurred. Studies of magnetic inclination measured in bore-core material also proved difficult to interpret [13] since definitive palaeomagnetic studies were not then available. A consequence of the new results reported here is that a framework is now provided within which the bore-core results can be interpreted and the results previously reported from South Australian formations assume new significance [15].

2. Geology of the sampling localities

Rock samples were collected from the Pound Quartzite (late Precambrian), Lower and Middle Cambrian and Upper Cambrian-Lower Ordovician formations in the northern Flinders Ranges, South Australia, and from the Lower Palaeozoic Tumblagooda Sandstone, Western Australia.

2.1. South Australia

The Adelaide Geosyncline, of which the Flinders Ranges comprise the northern part (Fig. 1), consists pri-

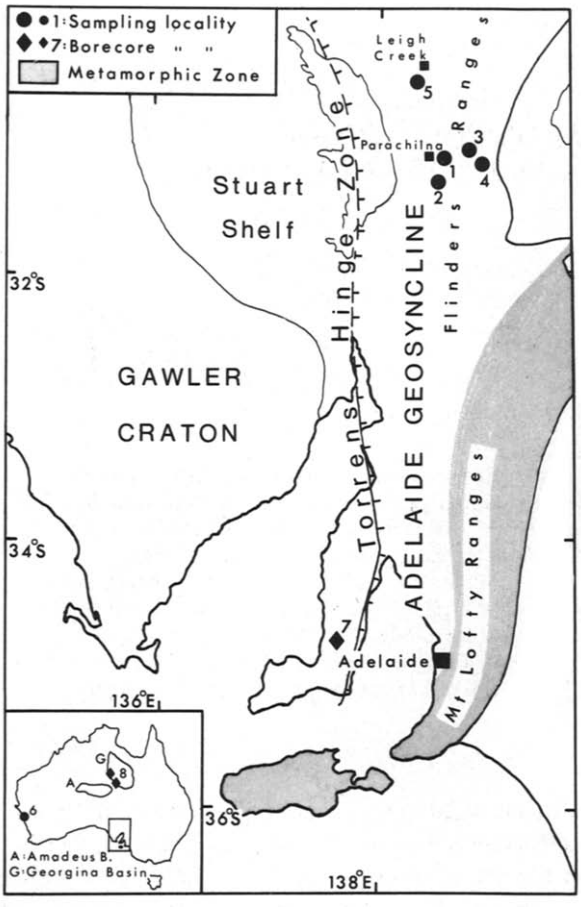


Fig. 1. The sampling localities in South Australia (full dots) are shown in relation to the principal structural elements of the Adelaide Geosyncline. 1,2,3 = Pound Quartzite; 5 = Lower Cambrian, 2,5 = Middle Cambrian (lower beds of the Lake Frome Group); 4 = Upper Cambrian–?Lower Ordovician (upper beds of the Lake Frome Group). Locality 6 represents the Tumblagooda Sandstone, Western Australia – it outcrops west of the Darling Fault, a major Late Jurassic structural line which parallels the west Australian coastline. The bore-core localities, referred to in the text, are marked as full diamonds, 7 = Lower Cambrian, Yorke Peninsula, and 8 = two Upper Cambrian bore sites (marked on the inset) in the Georgina Basin.

marily of a sequence of shallow marine and continental sediments of late Precambrian (Adelaidean System) and Cambrian age. The Gawler craton to the west was the major source of clastics during the depositional history of the geosyncline [16]. Sedimentation sporadically overlapped onto the eastern part of the craton (the Stuart Shelf) where thin, generally undisturbed, correlatives of parts of the Precambrian–Cambrian sequence occur. In the geosyncline zone, the sequence reaches a thickness in excess of 15 km. The sediments were folded during the Late Cambrian–Early Ordovician Delamerian Orogeny [17]. Radiometric ages obtained from metamorphosed shales and granites associated with the tectonic activity fall within the range 490–460 m.y. [18–21]. Effects of regional metamorphism are confined mainly to the southern and eastern regions of the orogen [22], the metamorphic grade decreasing in a northwesterly direction. In the northern Flinders Ranges the Upper Precambrian and Cambrian sediments are unmetamorphosed and preserved in open folds with predominantly north–south trending axes.

The Pound Quartzite, the youngest unit of the Wilpena Group, comprises two members; the lower red sandstone and siltone member was ampled for palaeomagnetic study. The upper member contains an Ediacaran fauna of latest Precambrian age [23]. It is overlain by thick sequences of Lower Cambrian Archaeocyathid limestones of the Hawker Group. The lower member was sampled in three localities (Fig. 1), (a) 22 samples were collected in Brachina Gorge at 5 sites spaced over 1000 m, the beds dip at 40° to the WNW, (b) 7 samples were collected from near the top of the member in Eregunda Creek where the beds dip at 35° to the NNE, and (c) 9 samples from the middle of the member were collected in Parachilna Gorge. The beds there dip at 70° to the SW

A notable feature of the Cambrian sequences in the Adelaide Geosyncline is the widespread occurrence of red-beds [16]. 15 samples were collected from the Lower Cambrian beds (dipping at 45° to the NE) in a 70 m long section near Aroona Dam (see Fig. 1). The Lower and Middle Cambrian beds at that locality are separated by a limestone which contains a Redlichia fauna. 29 samples representing the Middle Cambrian include 13 from Aroona Creek and 16 samples from a separate fold limb (dip 80° to the W) which outcrops over 200 m in Brachina Gorge. The Upper Cambrian sequence which represents the highest beds of the Lake Frome Group (and may be as young as Early Ordovician [24]) was sampled in Balcoracana Creek 11 samples were collected from a 150 m long section in beds dipping at 70° to the E.

2.2. Western Australia

On the eastern margin of the Carnarvon Basin a red-

bed sequence known as the Tumblagooda Sandstone outcrops The beds total more than 1800 m and rest with steep angular unconformity on Precambrian gneiss [25,26]. The Sandstone is overlain with slight unconformity by glacials which belong to the Lower Permian Lyons Group. Index fossils are absent, estimates of the age of the Tumblagooda Sandstone relying heavily on the presence of intertwined burrows and trails [27] Several geologists have considered the evidence relating to its age [25–28] it is now generally considered to be Ordovician although it may be as old as Middle Cambrian or as young as Early Silurian. 17 samples were collected from a coastal sequence near Bluff Point, about 10 km south of the Murchison River mouth (lat. 27 7°S, long 114.5°E). Above sea level, the sequence is about 150 m thick; sampling covered about 100 m. The beds are near-horizontal, and according to Condon [26] it is likely that the sequence was deposited with the dips currently displayed.

TABLE 1
Fold test data

	N*	k	
		uncorrected	corrected
Pound Quartzite			
NRM	30	6 0	3 2
cleaned	10	4 7	4.7
Lake Frome Group (Cm) (lower beds)			
NRM	29	4.3	1.4
cleaned	15	1 9	5.9

*N = number of samples, k = estimate of Fisher's precision parameter

3. Experimental procedure and results

All samples were collected as blocks and oriented using a sun compass of modified design [29]. Wherever possible, at least 3 specimens (28 X 28 mm cylinders) were cut from each block. Measurements of intensity and direction of magnetization were made using either a model SM1 P.A.R. spinner magnetometer or a „complete results" DIGIGO spinner magnetometer [30]. Thermal cleaning was carried out in air and in zero magnetic field using equipment similar to that described by McElhinny et al. [31]. Directions of magnetization were analysed using Fisher's statistics [32] and randomness of mean directions was tested with statistical methods described by Watson [33] and Vincenz and Bruckshaw [34].

3.1. Pound Quartzite

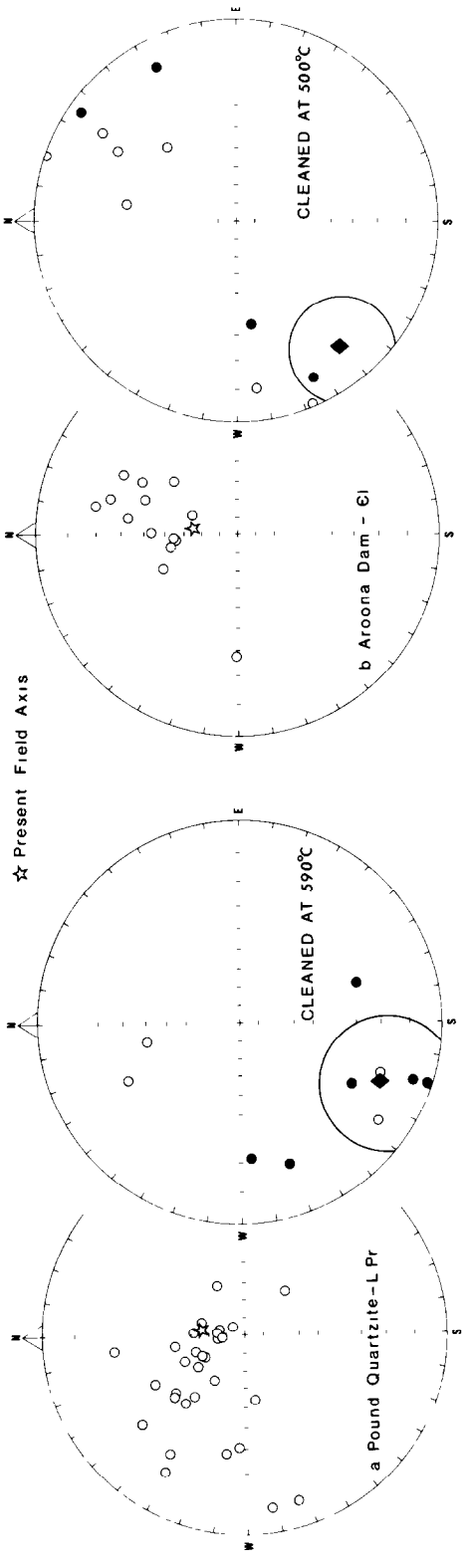
Intensities of magnetization of the samples were weak, ranging from 0.6 to 6.9 mA m^{-1}, with an arithmetic mean value of 2.0 mA m^{-1}. Mean directions of natural remanent magnetization (NRM) for samples considered non-random at the 95% probability level are shown in Fig. 2a prior to correction for bedding tilt. Application of Graham's fold test [35] increased the scatter (see Table 1). Pilot specimens were subjected to careful stepwise thermal demagnetization up to 670°C to test for the presence of a primary component of magnetization. Three types of behaviour were observed during those experiments.

(a) Directions initially oblique to the present field axis remained unchanged, their remanence was of the thermally discrete type [36].

(b) Specimen directions which were initially grouped around the present field axis underwent substantial angular changes at low temperatures and realized stable end-points in the temperature range 440–630°C. Their components of magnetic remanence were thermally distributed. We interpret these results to indicate the removal of a secondary viscous remanent magnetization at temperatures less than about 150–200°C (when a sharp drop in intensity occurred) and subsequent removal of a more stable secondary, possibly chemical, remanent magnetization.

(c) Specimen directions, which were initially grouped about the present field axis, remained unchanged. The highest blocking temperatures lay in the region 630–650°C. Within the practical limits of the investigation those specimens have probably lost their primary component of magnetization.

Although the pilot specimens were weakly magnetized after treatment (intensities < 1 mA m^{-1}), they indicated that a stable component of magnetization could b^ isolated in the temperature range 440–630°C. All remaining specimens were thermally cleaned at 590°C After bulk cleaning, within sample directions of 33 of


CLEANED AT 500°C
b Aroona Dam - €1
☆ Present Field Axis
CLEANED AT 590°C
a Pound Quartzite - L Pr
N
S
E
W

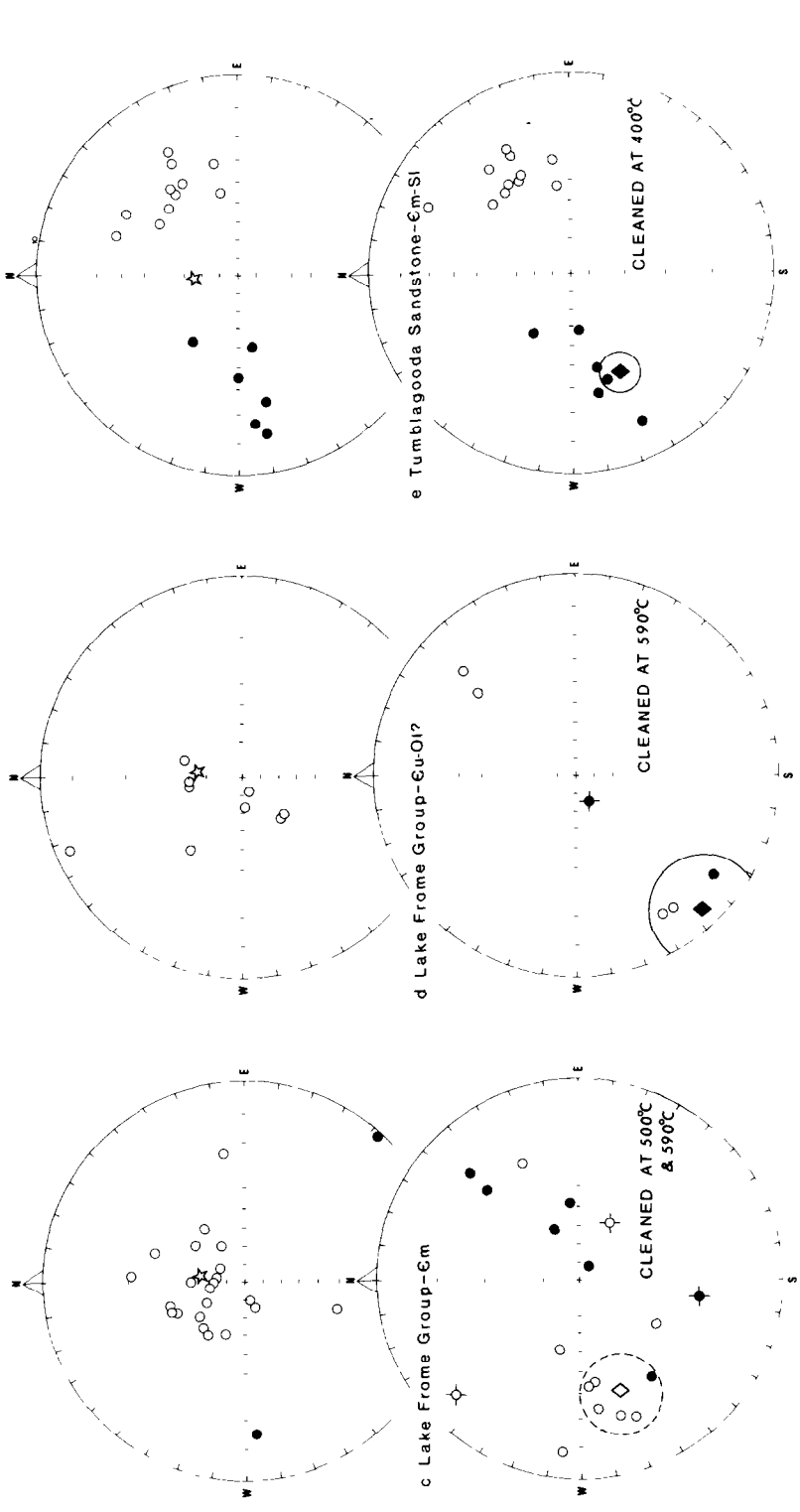


Fig. 2 Sample-mean directions of magetization for all age groups (dots) plotted on the Wulff equal angle stereographic projection. Full symbols represent positive (downward) directions and open symbols negative (upward pointing) directions. For each group of samples two plots are shown, one, NRM directions with respect to the present horizontal, includes the direction of the present geomagnetic field at the respective sampling locality and two, direction after treatment at temperatures indicated, and after bedding correction (where appropriate). The circles represent cones of confidence at the 95% probability level associated with each mean direction (indicated by a diamond). Crossed symbols represent sample-mean directions sufficiently oblique to the mean axes of magnetization that they may record intermediate field directions.

the 38 samples were considered random [34]. Closer examination of individual specimen directions from those 33 samples however, revealed that whilst some remained magnetized near the present field, others underwent systematic angular changes in excess of 90°. For 5 samples, it was possible to eliminate an aberrant specimen, leaving 2 specimens in which the primary component was present.

The 10 cleaned sample-means are shown in Fig. 2a after correction for bedding tilt; 2 samples are magnetised in the opposite sense. Comparison of directions before and after cleaning shows that they experienced an average systematic angular rotation of 80° away from the present field axis along an approximate N–S trend. The precision of the formation mean (see Table 1) was unchanged after correction for bedding tilt (before cleaning, this test relied heavily on samples from Eregunda Creek and Parachilna Gorge – most of those samples were rejected after treatment).

3.2. Cambrian sediments

NRM sample-mean directions of magnetization for the Early, Middle and Late Cambrian groups are plotted with respect to the present horizontal in Fig. 2b–d respectively. Their intensities lay in the range 0.5–33.0 mA m^{-1}. Directions in the Middle Cambrian sediments, with respect to the present horizontal, clustered around the present field axis, correction for bedding tilt increased the dispersion (see Table 1). Samples of Early and Late Cambrian age exhibited a planar distribution towards the present field axis. Preliminary experiments with pilot specimens indicated that stable components of magnetic remanence (regarded as primary) were isolated after partial thermal demagnetization at 500°C and 590°C. Following bulk treatment, results from 4 Lower Cambrian samples, 11 Middle Cambrian samples and 5 Upper Cambrian samples were rejected on the grounds that they either remained magnetized in a direction near the present field axis or that their remanence proved thermally unstable so resulting in low within-sample precision.

The remaining 11 sample-means of the Lower Cambrian sediments rotated systematically away from the present field by ca. 70° along an approximate NE–SW axis. Those directions are plotted with respect to the palaeohorizontal in Fig. 2b, they fall into two groups with opposite polarity.

Stable components of remanent magnetization were isolated in specimens from 18 samples of the Middle Cambrian sediments. 3 samples whose mean directions lay between 60° and 90° away from the formation mean were not included in the calculation of the pole position. Their directions were sufficiently oblique as to be considered possible records of genuine intermediate field directions. Fig. 2c shows the two groups with opposite polarities. After treatment the fold test was positive (see Table 1 and McElhinny [44]).

Results from the Upper Cambrian (?Lower Ordovician) sediments showed that only 6 samples responded succesfully to treatment. Directions in 5 samples indicate a reversal is present. One sample (see Fig. 2d) retained a rather steep inclination, it has been disregard ed for the purpose of calculating the formation mean direction. Whether it records an intermediate fiels direction is uncertain. Lower, Middle and Upper Cambrian pole positions are listed in Table 2.

3.3. Tumblagooda Sandstone

All samples apparently contained only one component of magnetization. The NRM directions plot in two, almost antiparallel, groups strongly oblique to the present field axis – see Fig. 2e. Pilot specimens, one from each sample, were heated in 100°C steps up to 600°C and at 650°C. No appreciable changes in their directions were produced below 600°C. At 650°C the scatter increased due to approaching the limit of the blocking temperature spectrum. In Fig. 3 the mean demagnetization–intensity curve obtained from the group of pilots with negative polarity shows a smooth decrease to 650°C. The mean curve obtained from the group of specimens with positive polarity firstly indicated an increase in intensity before decreasing smoothly to 650°C. It would appear that a very weak secondary component of magnetization was removed. The remaining specimens were measured after treatment at 400°C – the results are plotted in Fig. 2e. Samples from approximately the upper two-thirds of the sequence comprise the negative group of directions and samples from the lower one-third fall in the positive group. During sampling, estimates were made of the thicknesses of sediment separating the various sampled horizons. Using that field information it would seem that a field reversal occurs within about 2m. The formation mean direction of magnetization obtained after cleaning is given in Table 2.

The palaeomagnetic results described in this section

TABLE 2
Summary of the new palaeomagnetic results

Rock unit	Symbol	Age	N	R	Mean direction		South pole	A_{95}
					D	I		
South Australia								
Pound Quartzite	PQ	Latest Pr	10	8.091	203	17	6E 60S	23 5
Aroona Dam Sediments (Billy Creek equivalents?)	AD	€l	11	9.567	231	13	33E 36S	16 5
Lake Frome Group (lower)	LFG	€m	15	12.641	251	-29	25E 8S	14.5
Lake Frome Group (upper)	LFG	€u(?Ol)	5	4.775	227	6	26E 38S	12.0
Lake Frome Group (combined)	LFG	€m-u(?Ol)	20	16.508	243	-20	25E 16S	12.5
Western Australia								
Tumblagooda Sandstone	TS	O?(€m–Sl)	17	16.005	245	33	31E 30S	9 0

Notes (1) Age symbols : Pr = Precambrian, € = Cambrian, O = Ordovician; S = Silurian, D = Devonian, Upper, Middle and Lower divisions denoted by u, m and l.

(2) N = number of samples, R = length of resultant vector, D and I = mean declination and the inclination of remanent magnetization vector, and A_{95} = semi-angle of the cone of confidence at the 95% probability level (Fisher, 1953).

(3) The label „south pole" is assigned by continuity of these poles onto the Phanerozoic polar wandering curve for Australia.

demonstrate the presence of reversed polarities in each age group. This is taken as evidence that the magnetization, upon which the palaeomagnetic poles are based, is primary. In addition, the Middle Cambrian results from South Australia include a positive fold test (see Table 1) – the main phase of folding took place at the end of the Cambrian Period. Furthermore, the cleaned directions of magnetization are different from directions yet measured in younger Australian rock formations. The tropical palaeolatitudes indicated by the

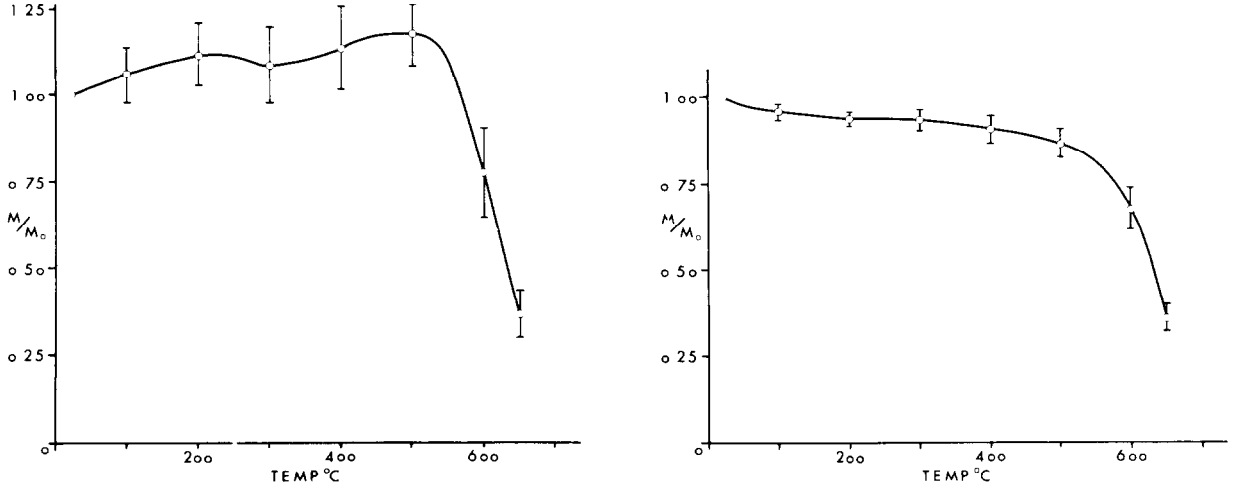


Fig. 3. Demagnetization–intensity curves for pilot specimens from the Tumblagooda Sandstone. The graph which indicates an initial increase of intensity with increasing treatment refers to specimens with positive polarities (see Fig. 2e). The graph which shows a smooth decrease of intensity from room temperature to 650°C refers to specimens with negative polarities. Error bars indicate the standard error of the mean intensity at each temperature.

results agree well with the palaeoclimatic data [37,38], e.g. the occurrence throughout the latest Precambrian and Cambrian of thick sequences of dolomites and limestones, some of which contain an Archaeocythid fauna [16]. The pole positions given in Table 2 are based on results which provide sufficient time-averages of the palaeomagnetic field such that secular changes are accounted for. In this context we considered it necessary to combine results from the upper and lower Lake Frome Group since only 5 samples from the younger section responded successfully to thermal treatment. The scatter of directions was frequently very high; Thompson [39] has previously commented upon similar problems and suggested a contributory cause particularly affecting Lower Palaeozoic rocks.

4. Discussion

Partial remagnetization is a problem often encountered during palaeomagnetic investigations. Rather than affecting preferred horizons (especially in sediments) degrees of remagnetization may be inhomogeneously distributed within individual beds and even, as in the case of the Pound Quartzite, within a single hand sample. Luck [7] encountered similar problems during his study of Lower Palaeozoic sediments from northern Australia. Also, during a study of the Permian red-beds from the Colorado Plateau, Farrell and May [40] rejected 90% of the specimens from their Monument Valley collection. The implication is that information relating to the primary magnetization might be retrievable if results are scrutinized at the specimen, rather than sample, level. This procedure was adopted with the palaeomagnetic data obtained by Briden [12, 13] from an earlier extensive study of South Australian late Precambrian and Cambrian formations. Giddings and Embleton [15] discuss those results more fully in terms of their particular relevance to the structural history of the Lower Palaeozoic in South Australia. Briden's material suffered severe remagnetization during the Late Cambrian–Early Ordovician orogenic activity and our interpretation of that data supports results presented here for the Late Cambrian–Early Ordovician of South Australia.

Briden also studied samples of sediments and tuffs belonging to the Dundas Group [12,41] and a syenite intrusion from Tasmania. They are Late Cambrian in age and were involved in the Late Cambrian Jukesian Orogeny [24]. Samples were subjected to laboratory stability tests – AF and thermal techniques. Upon re-examination of the data at the specimen level [15] we believe that a weak primary component has been re-

TABLE 3
Late Precambrian to Lower Palaeozoic palaeomagnetic poles used to define the polar wander curve with respect to the main Australian platform

Rock unit	Symbol	Age	South pole		A_{95}	Ref.
Northern Australia						
Antrim Plateau Volcanics	AV	Pr/Є1	9S	340E	17	[6]
Hudson Formation	HF	Єm	18N	19E	13	[7]
Jinduckin Formation	JF	Єu–Ol	13S	25E	11	[7]
Central Australia						
Arumbera Sandstone	AS	Pr/Є1	8N	325E	25	[8]
Hugh River Shale	HS	Є1–m	11N	37E	8	[8]
Stairway Sandstone	SS	Ol	2S	50E	8	[9]
Mereenie Sandstone	MS	S?–D	41S	40E	10	[9]
Tasmania						
Dundas Group	DG	Єu	23S	13E	11.5	[2,15]
Housetop Granite plus aureole	HG	Dm	67S	94E	27	[41]

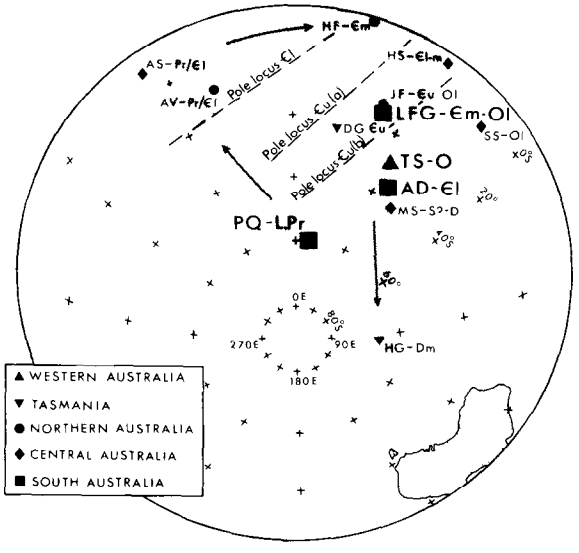


Fig. 4. The late Precambrian–Lower Palaeozoic apparent polar wander curve for the main Australian platform. Large symbols refer to palaeomagnetic poles obtained from studies described in this paper (see Table 2) and the small symbols refer to palaeomagnetic poles obtained from studies in other parts of Australia, presented previously (see Table 3). The Upper Cambrian pole (*DG*) obtained from the Dundas Group of Tasmania is also seen to lie on the curve in such a position that we need not invoke local rotation or translation of that region relative to the main Australian platform. For this reason and other evidence cited by McElhinny and Embleton [11], the Middle Devonian pole reported by Briden [41] from the Housetop Granite of Tasmania may also be regarded as relating to the main plattform (stippled) – the Precambrian outcrop in Tasmania is the most easterly occurrence in Australia. That pole tentatively allows us to extend the apparent polar wander curve into the Upper Palaeozoic. The dashed lines represent the loci of the poles obtained by Briden [13] from studies of bore-core material from Yorke Peninsula (Lower Cambrian) and the Georgina Basin (Upper Cambrian – *a* and *b*).

tained in specimens from 8 samples of the tuffs and syenite. They constitute two groups with opposite polarities and provide a positive fold test. The mean axis of magnetization corresponds to the direction of the Late Cambrian geomagnetic field defined from other regions of Australia. the south pole positions are given in Table 3 (see also Fig. 4).

The apparent polar wander curve for the Lower Palaeozoic now provides a suitable framework for the interpretation of results also obtained by Briden from investigations with bore-core material [13]. He calculated the colatitude (declination was indeterminate) of the sampling sites (the sites are shown in Fig.1). Arcs of three small circles relating to measurements from 2 bore-cores through Upper Cambrian sediments in the Georgina Basin (northern Australia) and a bore-core through Lower Cambrian sediments on Yorke Peninsula (South Australia) are drawn in Fig.4. They describe the loci of the palaeomagnetic poles inferred from colatitude estimates. These data are consistent with the rest of the Lower Palaeozoic results.

The Lower Cambrian pole (labelled *AD* in Fig.4) lies in correct chronological sequence with respect to the Pound Quartzite pole and the Arumbera Sandstone pole [8]. An Early Cambrian fossil found in the uppermost beds of the Arumbera Sandstone (central Australia) is also present in the Parachilna Formation [16,23]. That formation disconformably overlies the Pound Quartzite and is separated from the Aroona Dam sediments by several thousand metres of Archaeocyathid limestones belonging to the Lower Cambrian Hawker Group. However, the Aroona Dam pole apparently lies in an anomalous position when compared with the pole yielded by the Hudson Formation. Samples from the South Australian sediments were collected just below a thin Redlichia bearing limestone (separating them from the Lake Frome Group) and the Hudson Formation was sampled above the main Redlichia horizon in the Negri Group, Northern Territory [7]. At present we have no explanation for this but await additional palaeomagnetic and palaeontological evidence which may help resolve the question.

The results from the Pound Quartzite allow us to extend the apparent polar wander path into the latest Precambrian (see Fig. 4). The next youngest group of poles belong to the Arumbera Sandstone and the Antrim Plateau Volcanics [6]. Initial deposition of the Arumbera Sandstone occurred during the final stages of the Petermann Ranges Orogeny and continued when orogenic activity ceased [42]. Daily et al [24] regard the principal phase of that orogeny as post-Ediacaran, i.e. the fauna preserved in the upper member of the Pound Quartzite. It is therefore older than the Arumbera Sandstone and that it could be substantially so is suggested by (a) the presence of an orogeny between the two, and (b) the Pound Quartzite pole was derived from the pre-Ediacaran lower member.

Results reported here from South Australia and Western Australia provide independent support for

the polar wander curve previously considered common only to northern and central Australia. They are compatible with the main platform having maintained its physical integrity since late Precambrian time. During the period latest Precambrian to Lower Palaeozoic the apparent polar wander curve describes a loop of approximately 180°. Events such as the late Palaeozoic Alice Springs Orogeny probably result from intraplate compression, the effects of which are not detectable palaeomagnetically. Southeastern Australia (excluding Tasmania) which is apparently not underlain by Precambrian rocks [43], has however suffered a relatively complex tectonic history, partly decipherable with the palaeomagnetic method [10].

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